

BAJARI (PEARL MILLET)

Complete Deep Farming, Grading & Traceability Report

Maharashtra Edition

Districts Covered: Dhule · Nandurbar · Nashik · Akola · Beed · Ahilyanagar · Chhatrapati Sambhajnagar

*Compiled from ICAR Millets Development Directorate, Government of India Official Records,
Maharashtra State Agriculture Department & Millets Now Traceability Platform V2*

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1. INTRODUCTION & ORIGIN OF BAJARI MILLET

Bajari, commonly known as Pearl Millet and botanically identified as *Cenchrus americanus* (syn. *Pennisetum glaucum*), is one of the most drought-resilient cereal crops cultivated in India. It originated in the Sahel region of West Africa and moved into the Indian subcontinent in prehistoric times, where it became deeply embedded in dryland food systems.

In Maharashtra, Bajari is both a food-security crop and a climate-risk management crop. It performs under low and erratic rainfall, produces dependable fodder, and provides grain for bhakri, flour, malted foods, porridges and modern millet products.

1.1 Global & National Standing

Pearl millet is the most widely grown millet globally and India remains the largest producer and consumer. It is cultivated across Rajasthan, Maharashtra, Gujarat, Haryana, Uttar Pradesh, Karnataka, Madhya Pradesh and parts of peninsular India.

Nationally, Bajari is valued for its short duration, low water requirement and adaptability to sandy loam, medium black and light textured soils. Maharashtra is a significant producer, particularly in Khandesh, Marathwada, Vidarbha and rainfed belts adjoining Nashik and Ahilyanagar.

1.2 Why Bajari Millet Matters

- Climate resilience: survives intermittent dry spells and high temperature better than many cereals.
- Nutritional value: rich in iron, energy, protein, minerals and slow-release carbohydrates.
- Dual-purpose value: provides both grain and nutritious dry fodder for cattle, goats and sheep.
- Short crop cycle: most improved varieties mature in 75–95 days, matching Maharashtra kharif rainfall windows.
- Market relevance: demand is rising for millet flour, ready mixes, flakes, extruded snacks and health foods.

1.3 Agronomic Significance in Maharashtra

Bajari fits the semi-arid ecology of Dhule, Nandurbar, Beed, Akola, Nashik, Ahilyanagar and Chhatrapati Sambhajinagar. It uses conserved soil moisture efficiently and can recover after short moisture stress when crop population is maintained.

Its role in Maharashtra farming systems extends beyond grain: stalks support livestock feed security, residues improve soil cover, and intercropping with pigeon pea, green gram, cowpea or sesame spreads production risk.

Vernacular Name	Language/State	Vernacular Name	Language/State
Bajari / Bajra	Marathi, Hindi, Gujarati	Sajje	Kannada
Kambu / Cumbu	Tamil, Malayalam	Sajjalu	Telugu
Bajra	North India	Pearl Millet	English trade name

2. NUTRITIONAL PROFILE & HEALTH BENEFITS

Pearl millet is an energy-dense nutri-cereal. Its nutritional profile supports food security in dryland regions and creates value in health-oriented markets because it is naturally gluten-free and contains useful levels of iron, minerals and dietary fibre.

2.1 Nutritional Composition per 100 g (Dry Grain)

Nutrient	Quantity	% Daily Value / Remark
Energy	345–370 kcal	Dense energy for dryland diets
Carbohydrates	62–70 g	Slow-release starch; supports satiety
Protein	10–12 g	Higher than rice; useful cereal protein
Dietary Fibre	1.2–11.5 g	Varies by source, processing and variety
Fat	4–6 g	Higher lipid fraction than many cereals
Iron	5–16.9 mg	Strong micronutrient advantage; biofortified lines add value
Calcium	38–42 mg	Moderate mineral contribution
Phosphorus	240–300 mg	Supports bone and energy metabolism
Magnesium	120–140 mg	Useful for nerve and muscle function
Zinc	2.5–3.5 mg	Important for immunity and growth
Polyphenols	Moderate	Antioxidant potential; may reduce digestibility if excessive

2.2 Key Health Benefits

- Gluten-free grain suitable for wheat-free food formulations.
- High iron content supports nutrition programmes aimed at reducing anaemia risk.
- Higher fibre and slowly digestible starch help create low-to-moderate glycaemic food products when minimally processed.
- Protein, minerals and energy make it suitable for school meals, community nutrition and labour-intensive rural diets.
- Pearl millet flour, when fresh and properly stored, is useful in bhakri, porridges, malt mixes and composite bakery blends.

3. TYPES OF SEEDS & SEED SELECTION

Seed selection determines crop uniformity, disease risk, flowering synchrony and final market grade. Maharashtra farmers should select seed according to rainfall, soil type, maturity duration, fodder need and the end-use market.

Seed Type	Description	Advantages	Limitations
Hybrid Seeds	F1 seed produced from selected parental lines.	High yield potential, uniform earheads, good response to fertilizer.	Seed should not be saved for next season; higher purchase cost.
Certified Seeds	Seed meeting official purity and germination standards.	Reliable germination, traceable lot, known variety performance.	Availability can be seasonal; must be sourced early.
Biofortified Seeds	Varieties bred for improved iron and zinc.	Premium nutrition branding; useful for institutional procurement.	Needs identity preservation and traceability.
Local / Farm-saved Seeds	Seed retained from farmer-selected crop.	Low cost and locally adapted.	Variable purity, disease carry-over, lower grade consistency.

Recommendation: Use certified or truthfully labelled seed of locally recommended hybrids/varieties with germination above 85%, physical purity above 98%, and no visible fungal contamination. Retain seed-lot number in the traceability record.

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4. SEED VARIETY DEEP ANALYSIS

Pearl millet variety choice should balance maturity, drought tolerance, downy mildew resistance, grain colour, fodder yield and market preference. In Maharashtra, short to medium duration varieties are preferred for kharif rainfall uncertainty, while irrigated summer Bajari requires heat-tolerant hybrids.

4.1 Variety Performance Matrix

Variety / Hybrid	Duration	Yield Potential	Disease Profile	Suitable Districts	Special Position
ICTP 8203 / Dhanshakti	80–85 days	Good grain; high iron trait	Moderate resistance; monitor downy mildew	Dhule, Nandurbar, Beed, Akola	Biofortified nutrition positioning and institutional market appeal.
HHB 67 Improved	65–75 days	Medium; reliable under stress	Downy mildew resistant line	Low rainfall pockets of Beed, Dhule	Very short duration escape mechanism for delayed rains.
GHB 558 / GHB group	80–90 days	High under good management	Good disease tolerance	Nashik, Ahilyanagar, Chhatrapati Sambhajnagar	Useful where rainfall is moderate and fertilizer response is possible.
MBBH / State hybrids	75–90 days	High grain and fodder	Select locally notified lots	Khandesh and Marathwada	Choose only locally tested seed lots from reliable suppliers.
Local white/grey landraces	85–100 days	Variable	Variable; may harbour seed-borne disease	Traditional pockets	Useful for taste and local bhakri markets when cleaned and graded.

For premium traceable grain, variety identity must be recorded at sowing and preserved through harvest, threshing, storage and sale. Mixed lots reduce grade consistency and weaken buyer confidence.

4.1 Variety Performance Matrix — Detailed Field Notes

Selection Factor	Preferred Bajari Choice	Reason for Maharashtra Conditions
Delayed monsoon	HHB 67 Improved or other short-duration line	Escapes terminal drought and matures within a compressed kharif window.
Premium nutrition market	ICTP 8203 / Dhanshakti type biofortified line	Iron-rich identity can be preserved through traceable aggregation.
Fodder priority	Dual-purpose local/state hybrid with strong stover	Livestock feed value is important in Dhule, Nandurbar, Beed and Ahilyanagar.
Disease-prone humid pocket	Downy mildew tolerant hybrid	Prevents systemic infection and malformed earheads.
Processor contract	Uniform grain colour and size	Milling recovery and flour colour consistency improve buyer acceptance.

- Do not mix hybrid seed lots in the same plot when the harvest is intended for branded flour.
- Where local taste preference is important, compare grain colour and bhakri texture before bulk adoption.
- Use village-level demonstrations for one season before shifting an entire FPO cluster to a new hybrid.
- Maintain a seed packet photograph, bill and lot number in the digital traceability record.

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5. SOIL REQUIREMENTS & PHYSICAL PROPERTIES

5.1 Ideal Soil Types

Bajari prefers well-drained sandy loam to medium black soils with good aeration. It can tolerate lower fertility better than maize, but responds strongly to balanced fertilizer and organic matter. Waterlogging is highly damaging, especially during germination and early vegetative growth.

Soil Type	Suitability	Management Note
Sandy loam	Excellent	Best for drainage; requires split nitrogen and moisture conservation.
Medium black soil	Good	Useful where drainage is adequate; avoid compaction.
Light red / gravelly soil	Moderate	Use FYM, ridge-furrow planting and drought-tolerant varieties.
Heavy waterlogged clay	Poor	Avoid for grain crop; high risk of seedling mortality and root disease.

5.2 Critical Soil Parameters

Parameter	Ideal Range	Field Interpretation
pH	6.0–8.0	Tolerates mild alkalinity; extreme salinity reduces emergence.
Electrical Conductivity	< 2 dS/m	Higher EC reduces stand and tillering.
Organic Carbon	> 0.5%	Improves water holding and nutrient buffering.
Drainage	Rapid to moderate	No standing water after heavy rain.
Available N	Low to medium common	Nitrogen is usually the key yield-limiting nutrient.
Available Zn	Site-specific	Apply zinc sulphate where soil test indicates deficiency.

5.3 Land Preparation Protocol

- One deep summer ploughing to expose pests and break hard pans where feasible.
- Two harrowings before sowing to create fine tilth and conserve early monsoon moisture.
- Use broad-bed furrow or ridge-furrow on medium black soils to remove excess rainwater.
- Apply well-decomposed FYM during final land preparation and incorporate evenly.

5.3 Land Preparation Protocol — Field Diagnostic Checklist

Field Observation	Likely Impact	Corrective Action
Hard crust after first rain	Patchy germination and weak seedlings.	Use light harrowing or crust breaking before seedlings are damaged.
Water stands for more than 24 hours	Root injury, seedling rot and poor tillering.	Open field drains or shift to ridge-furrow layout in future seasons.
Low organic matter and pale crop	Low water holding and nitrogen stress.	Add FYM/compost and split nitrogen after rainfall.
Termite or white grub history	Stand loss after sowing.	Deep summer ploughing and local plant-protection advisory.
Uneven slope	Runoff, soil erosion and non-uniform crop.	Contour sowing, compartment bunding and runoff checks.

For traceable production clusters, field readiness should be verified before seed distribution. A simple checklist with soil type, drainage, previous crop, FYM application and sowing method prevents avoidable variation between farmer lots.

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6. CLIMATE REQUIREMENTS & WATER MANAGEMENT

6.1 Ideal Climate

Climate Parameter	Optimal Range / Details
Temperature	25–35°C during vegetative growth; warm weather at flowering.
Rainfall	350–600 mm well distributed; crop can survive lower rainfall with reduced yield.
Photoperiod	Most improved cultivars are day-neutral or weakly sensitive.
Humidity	Low to moderate; prolonged humid weather favours downy mildew and blast.
Critical stress stages	Seedling establishment, booting, flowering and grain filling.

6.2 Irrigation Management — Stage-wise Protocol

Growth Stage	Days After Sowing	Irrigation Need	Critical Priority
Germination	0–7 DAS	Ensure moist seed zone; avoid crusting.	High
Tillering	20–30 DAS	Life-saving irrigation if dry spell exceeds 10–12 days.	Medium
Booting / Flowering	40–55 DAS	Most critical irrigation for grain set.	Very High
Grain Filling	60–75 DAS	Light irrigation improves grain boldness where available.	High
Maturity	80+ DAS	Stop irrigation; allow drying before harvest.	Low

6.3 Drought Management Techniques

- Select short-duration hybrids when monsoon onset is delayed beyond the normal sowing window.
- Maintain optimum plant population; gap-fill early if emergence is patchy.
- Use intercultivation to break soil crust and create dust mulch after rainfall.
- Apply FYM or compost to improve moisture retention in light soils.
- Use contour cultivation, tied ridges or compartment bunding on sloping dryland fields.

6.3 Drought Management Techniques — Stress Response Table

Crop Stage	Visible Stress Symptom	Immediate Response	Quality Risk if Ignored
Emergence	Patchy stand and curled seedlings.	Gap-fill within 7–10 days and break crust.	Low population and mixed maturity.
Tillering	Leaf rolling at noon; slow growth.	Interculture to conserve moisture; avoid heavy top dressing without rain.	Reduced earhead number.
Booting	Delayed ear emergence and pale canopy.	Give protective irrigation where available.	Poor earhead size and low grain set.
Flowering	Stigma drying, poor pollination.	Life-saving irrigation and avoid spray stress.	Shriveled grain and grade loss.
Grain filling	Early drying of leaves and light grains.	Protect residual moisture; avoid late harvest delay.	Low test weight and higher immature grains.

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7. SOWING MANAGEMENT

7.1 Time of Sowing

Kharif sowing should begin after receipt of effective monsoon rainfall sufficient to wet the seed zone. In most Maharashtra Bajari belts, the practical sowing window is mid-June to mid-July; delayed sowing requires short-duration varieties. In Marathwada, dry sowing just before dependable monsoon showers is practiced in some locations under local advisories.

7.2 Sowing Parameters

Parameter	Recommendation
Seed rate	3–4 kg/ha for line sowing; 4–5 kg/ha where stand risk is high.
Spacing	45 cm row spacing × 10–15 cm plant spacing for rainfall >450 mm zones.
Depth	2.5–4 cm; deeper sowing delays emergence and weakens seedlings.
Plant population	1.75–2.0 lakh plants/ha in normal rainfall zones.
Method	Seed drill preferred; maintain uniform depth and row spacing.

7.3 Intercropping Systems (ICAR Recommended)

System	Row Ratio	Purpose
Bajari + Pigeon pea	2:1 or 4:2	Risk diversification and pulse income.
Bajari + Green gram	2:1	Short-duration legume and soil nitrogen contribution.
Bajari + Cowpea	2:1	Fodder plus grain, improves soil cover.
Bajari + Sesame	3:1	Low water companion crop for dry belts.

8. SEED TREATMENT PROTOCOL

Seed treatment protects the initial stand and reduces avoidable disease carry-over. Use locally approved products and doses as advised by agriculture officers; do not mix incompatible chemicals and bioagents in the same step.

Treatment Step	Material / Practice	Purpose
Seed cleaning	Remove broken, shrivelled and discoloured seed.	Improves germination and uniformity.
Fungicidal treatment	Treat with recommended fungicide such as thiram/carbendazim class as per label.	Controls seed-borne fungi and seedling rot.
Biofertilizer coating	Azospirillum + PSB after chemical treatment has dried.	Improves nitrogen and phosphorus use efficiency.
Downy mildew care	Use resistant variety and clean seed; follow state advisory for metalaxyl-based treatment where recommended.	Reduces early systemic infection.
Traceability record	Record seed lot, treatment material, dose, date and operator.	Creates audit-ready sowing record.

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9. COMPLETE CROP CYCLE — STAGE-WISE GUIDE

Stage	Days	Field Operations	Quality Objective
Pre-sowing	May–June	Soil test, FYM application, land levelling, seed procurement.	Prepare uniform field and verified seed source.
Sowing & emergence	0–10 DAS	Line sowing, seed treatment, light crust breaking if needed.	Uniform plant stand.
Early vegetative	10–25 DAS	Gap filling, thinning, first weeding/interculture.	Optimum population and reduced weed competition.
Tillering	25–40 DAS	Top-dress nitrogen, interculture, monitor downy mildew.	Strong tillers and healthy foliage.
Booting & flowering	40–60 DAS	Life-saving irrigation if possible, pest monitoring.	High grain set and earhead development.
Grain filling	60–80 DAS	Protect from bird damage; avoid late nitrogen.	Bold, well-filled grains.
Maturity	80–95 DAS	Harvest at physiological maturity when earheads dry.	Low moisture and minimal shattering.
Post-harvest	After harvest	Drying, threshing, cleaning, grading, storage.	Marketable Grade A or B lot.

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10. DISEASE & PEST MANAGEMENT

Bajari is hardy but can suffer major losses from downy mildew, blast, smut, ergot and insect pests under favourable conditions. Integrated management must begin with resistant seed and clean field practices.

10.1 Stage-wise Disease Profile

Disease / Pest	Likely Stage	Symptoms	Severity Risk	Control Focus
Downy mildew	Seedling to vegetative	Chlorotic leaves, white downy growth, malformed earheads.	High in humid pockets	Resistant hybrid, seed treatment, rogue infected plants.
Blast	Vegetative to grain filling	Spindle-shaped leaf lesions and neck infection.	Medium to high	Avoid dense canopy; use tolerant varieties and advisory-based spray.
Ergot	Flowering	Honeydew droplets, sclerotia replacing grains.	High in wet flowering	Synchronised flowering, remove infected earheads, clean seed.
Smut	Earhead emergence	Sori replacing grains in earhead.	Medium	Seed treatment and resistant varieties.
Shoot fly / stem borer	Seedling to vegetative	Dead hearts, weak tillers.	Localised	Timely sowing, field sanitation, need-based control.
Bird damage	Grain filling	Missing grains and broken earheads.	High near villages/trees	Community scaring and timely harvest.

10.2 Integrated Disease Management (IDM)

- Use resistant/tolerant varieties and avoid repeated use of susceptible hybrids in disease-prone villages.
- Treat seed and record the treatment in the traceability log.
- Maintain recommended spacing; dense crop canopies increase humidity and disease spread.
- Rogue downy mildew infected plants early and remove ergot/smut earheads before threshing.
- Rotate with pulses or oilseeds; avoid continuous Bajari in the same field where disease pressure is high.

10.2 Integrated Disease Management (IDM) — Monitoring Calendar

Monitoring Window	What to Inspect	Action Threshold / Action
7–15 DAS	Emergence, seedling rot, early downy mildew.	Rogue infected plants; verify seed treatment records.
20–35 DAS	Tillering, leaf spots, shoot fly dead hearts.	Field sanitation and need-based advisory control.
40–55 DAS	Booting, blast lesions, ergot-risk weather.	Avoid dense humidity; prepare for flowering surveillance.
Flowering	Honeydew droplets, smut balls, insect activity.	Remove infected earheads before they contaminate the lot.
Pre-harvest	Bird damage, mould, off-type earheads.	Harvest clean plots separately and keep grade-risk lots apart.

Digital field photos should be captured at tillering, flowering and pre-harvest stages. This gives buyers confidence that the lot was monitored and that disease-affected earheads were removed before threshing.

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11. FERTILIZER MANAGEMENT — NPK, ORGANIC & MICRO-NUTRIENTS

11.1 NPK Recommendations (ICAR — Pearl Millet)

Pearl millet responds well to balanced nutrition. Exact fertilizer must be based on soil test, rainfall zone and whether the crop is rainfed or irrigated. The following schedule is a practical Maharashtra dryland reference.

Condition	Nitrogen	Phosphorus	Potassium	Remarks
Rainfed low fertility	40–60 kg N/ha	20–30 kg P ₂ O ₅ /ha	Soil-test based	Apply part basal and part at 25–30 DAS if moisture is available.
Assured rainfall / better soils	60–80 kg N/ha	30–40 kg P ₂ O ₅ /ha	20 kg K ₂ O/ha if deficient	Higher response where plant stand is uniform.
Irrigated / summer	80–100 kg N/ha	40 kg P ₂ O ₅ /ha	20–40 kg K ₂ O/ha	Split nitrogen to reduce loss and lodging risk.

11.2 Fertilizer Application Schedule

Timing	Application	Operational Note
Basal	Full phosphorus + potassium and 50% nitrogen.	Place away from seed because pearl millet seed is sensitive to fertilizer burn.
25–30 DAS	Remaining nitrogen top dressing.	Apply after weeding and when soil moisture is adequate.
Deficiency correction	Zinc sulphate/ferrous sulphate where soil test or symptoms indicate.	Use state advisory dose; avoid blind micronutrient application.

11.3 Organic Fertilizer Options

- FYM or compost: 5–10 t/ha incorporated before sowing, especially on light soils.
- Vermicompost: useful in seed rows or high-value traceable plots where cost allows.
- Green manuring or legume intercrop residues: improves soil organic matter over seasons.
- Biofertilizers: Azospirillum and PSB can reduce chemical fertilizer dependence when used correctly.

11.3 Organic & Micro-Nutrient Field Indicators

Symptom	Probable Cause	Corrective Management
Uniform pale green crop	Nitrogen deficiency or low mineralisation.	Top-dress nitrogen after weeding when soil moisture is adequate.
Purple tinge in young plants	Phosphorus stress in cool or low-P soil.	Use soil-test phosphorus and place basal fertilizer correctly.
Interveinal chlorosis on young leaves	Iron/zinc stress, often in calcareous soil.	Use soil-test based micronutrient correction or advisory spray.
Weak stalks and lodging	Excess nitrogen, dense crop or potassium imbalance.	Avoid late nitrogen; keep recommended spacing.
Low fodder weight	Poor organic matter and moisture stress.	Add FYM/compost and improve moisture conservation over seasons.

Nutrient decisions should be recorded as product, dose, date and field condition. Traceability does not require excessive inputs; it requires justified, auditable input use.

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12. GRAIN GRADING SYSTEM — A, B & C GRADES

12.1 Grade Parameters (Government Reference Standard)

The A/B/C framework below is aligned to Indian trade practice using FSSAI/AGMARK quality factors: moisture, foreign matter, other edible grains, damaged grains, immature/shrivelled grains, weevilled grains, odour, live insects and overall wholesomeness.

Parameter	Grade A	Grade B	Grade C
Moisture	≤ 12.0%	≤ 14.0%	≤ 16.0%
Foreign matter	≤ 0.5%	≤ 1.0%	≤ 2.0%
Other edible grains / admixture	≤ 1.0%	≤ 2.0%	≤ 3.0%
Damaged / discoloured grains	≤ 1.5%	≤ 3.0%	≤ 5.0%
Immature / shrivelled grains	≤ 3.0%	≤ 5.0%	≤ 8.0%
Live insects / musty odour	Absent	Absent	Rejected if present
Best market use	Food processing, premium flour, institutional procurement	General food grain and local mills	Feed, industrial use or re-cleaning

12.2 Grade-wise Visual Characteristics

- Grade A: bold, uniform grey-green or cream grains; clean lot; low broken grain and no smell.
- Grade B: usable grain with slight size variation and limited immature or discoloured grains.
- Grade C: visibly mixed, higher shrivelling, broken grains or dusty lots requiring cleaning.

12.3 Achieving Grade A — Key Practices

- Harvest only after physiological maturity and dry earheads before threshing.
- Thresh on clean tarpaulin or concrete floor; avoid soil contact.
- Clean with sieve/aspirator and remove light, immature and foreign material.
- Dry grain to 11–12% moisture before bagging.
- Store in clean, dry bags with lot ID, variety, village, harvest date and moisture record.

12.1 Grade Parameters — Sampling & Assaying Protocol

Step	Protocol	Traceability Record
Lot definition	Keep one variety, one village/cluster and one harvest window together.	Lot ID, farmer IDs and bag count.
Primary sampling	Draw small samples from multiple bags, not only the top layer.	Sampler name, date and bag numbers.
Composite sample	Mix and reduce sample using clean tray or divider.	Composite sample code.
Moisture testing	Use calibrated moisture meter or lab method.	Reading, device ID and operator.
Physical purity	Measure foreign matter, damaged, immature and other grains.	Assay sheet and grade decision.
Final sealing	Tag bags after grade approval.	QR code and dispatch record.

A clean sampling protocol prevents disputes between farmers, FPOs, processors and buyers. The same table can be used by Millets Now field officers as the minimum quality-assurance record for Grade A and Grade B Bajari lots.

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13. DISTRICT-WISE DEEP FARMING ANALYSIS — MAHARASHTRA

13.1 Dhule District

Parameter	District Guidance
Climate / Soil	Semi-arid Khandesh belt with strong Bajari suitability, medium to light soils and frequent dry spells.
Recommended Practice	Kharif after effective monsoon; use drought-tolerant hybrids and maintain 45 cm rows.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Bajari + pigeon pea or green gram; grade focus on drying and dust removal.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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13.2 Nandurbar District

Parameter	District Guidance
Climate / Soil	Tribal and hilly pockets with variable rainfall, light soils and traditional millet food culture.
Recommended Practice	Use locally accepted varieties, contour cultivation and community seed sourcing.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Traceability can highlight tribal origin, low-input cultivation and bhakri grain.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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13.3 Akola District

Parameter	District Guidance
Climate / Soil	Vidarbha dryland conditions with medium black soils and rainfall variability.
Recommended Practice	Use drainage care in black soils; avoid water stagnation after heavy rain.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Intercrop with pigeon pea or sesame; strict moisture management before storage.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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13.4 Beed District

Parameter	District Guidance
Climate / Soil	Drought-prone Marathwada zone where short-duration Bajari is a risk-reduction crop.
Recommended Practice	Prefer early maturing hybrids when rains are delayed; life-saving irrigation at flowering if possible.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Grade losses commonly arise from shrivelling; focus on plant population and timely harvest.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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13.5 Chhatrapati Sambhajnagar District

Parameter	District Guidance
Climate / Soil	Semi-arid Marathwada district with mixed black and light soils.
Recommended Practice	Use ridge-furrow on heavier soils and balanced fertilizer where rainfall permits.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Good scope for traceable grain aggregation through farmer groups.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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13.6 Nashik & Ahilyanagar Districts

Parameter	District Guidance
Climate / Soil	Rainfed to semi-irrigated belts with market connectivity and diverse cropping systems.
Recommended Practice	Use high-yielding hybrids in assured rainfall pockets; manage blast in humid spells.
Intercrop / Rotation	Pigeon pea, green gram, cowpea, sesame or fallow rotation depending on rainfall and market.
Water Requirement	Mostly rainfed; one protective irrigation at booting/flowering can significantly improve grain filling.
Grading & Market Note	Strong opportunity for premium flour, retail packs and value-added products.

- Use certified seed distributed before monsoon onset and record seed-lot identity.
- Aggregate clean grain only after moisture testing and visual grade confirmation.
- Keep village-wise lots separate where soil type, rainfall or variety differs materially.

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14. HARVESTING, THRESHING & STORAGE

14.1 Harvesting Guidelines

- Harvest when earheads turn dry and grains are hard, usually 80–95 days after sowing depending on variety.
- Cut earheads first for grain quality; harvest stalks later for fodder when appropriate.
- Avoid harvest immediately after rain because wet earheads increase mould risk.

14.2 Threshing

- Dry earheads on clean floor or tarpaulin before threshing.
- Use mechanical thresher at suitable speed to reduce broken grains.
- Winnow and sieve immediately to remove dust, husk, immature grains and stones.

14.3 Storage Best Practices

Storage Factor	Best Practice
Moisture	Store at 11–12% for premium lots; never bag warm grain.
Bags	Use clean jute/PP bags; label with lot ID and date.
Structure	Dry, ventilated godown with pallets and wall clearance.
Inspection	Check monthly for moisture migration, insects, rodents and odour.
Traceability	Do not mix varieties, villages or harvest dates without recording blending.

15. INPUT CYCLE & COST STRUCTURE

15.1 Complete Input Cycle

Input / Operation	Timing	Typical Requirement	Cost Sensitivity
Seed	Before sowing	3–4 kg/ha certified seed	High for hybrids; affects yield and uniformity.
FYM / Compost	Land prep	5–10 t/ha where available	Transport is major cost.
Fertilizer	Basal + 25–30 DAS	NPK as per soil test	Nitrogen split depends on rainfall.
Biofertilizer	Seed treatment	Azospirillum + PSB	Low cost; quality of culture matters.
Weeding / Interculture	15–35 DAS	One or two operations	Critical because early weeds reduce yield sharply.
Plant protection	Need-based	Disease/pest advisory based	Avoid unnecessary sprays; preserve audit record.
Harvest & threshing	Maturity	Labour or machine	Clean threshing improves grade premium.
Cleaning & packaging	Post-harvest	Sieve, moisture test, bags, tags	Essential for Grade A/B marketing.

Cost varies by district, seed type, labour rate, irrigation access and mechanisation. For traceable lots, the additional cost of moisture testing, lot tagging, clean tarpaulin drying and digital entry is small compared with the value of buyer confidence.

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16. MILLETS NOW — DIGITAL TRACEABILITY PLATFORM (V2)

16.1 Platform Vision & Goals

- Build a verifiable Bajari grain identity from seed purchase to final buyer.
- Enable farmer groups, FPOs and processors to document quality, origin and production practices.
- Support premium markets through lot-level records, QR codes and tamper-resistant audit trails.

16.2 End-to-End Traceability Workflow

- Farmer registration and plot geo-tagging.
- Seed lot, variety, sowing date and seed treatment entry.
- Crop-stage observations for rainfall, fertilizer, irrigation and pest management.
- Harvest, drying, threshing, cleaning and moisture testing records.
- Grading, bagging, warehouse entry, buyer dispatch and QR-based verification.

16.3 System Architecture Layers

Layer	Components
Farmer Application	Profile, farm map, crop calendar, advisory alerts and input records.
Field Officer Console	Inspection checklist, photo evidence, crop-stage approval and grade validation.
Quality Layer	Moisture, foreign matter, damaged grains, shrivelled grains and lot sampling records.
Blockchain Layer	Lot hash, timestamp, transaction ID, custody transfer and immutable audit record.
Buyer Interface	QR verification, certificate view, origin map and quality summary.

16.4 Blockchain Traceability Record — Data Fields

Data Field	Example Entry
Crop	Bajari / Pearl Millet
Botanical name	Cenchrus americanus / Pennisetum glaucum
Farmer ID / FPO ID	MN-MH-DHL-000124
District / Village	Dhule / village name
Variety / Seed lot	Dhanshakti / certified seed lot number
Sowing and harvest date	15 June 2026 / 12 September 2026
Moisture and grade	11.8% / Grade A
Bag IDs	MN-BJ-2026-0001 to 0050
Hash / QR status	Active verification link

16.2 End-to-End Traceability Workflow — Quality Gates

Gate	Verification Item	Approval Output
Seed gate	Seed invoice, variety, lot number and treatment record.	Seed lot approved for sowing.
Field gate	Geo-tag, sowing date, crop-stage photos and input entries.	Field record locked for crop season.
Harvest gate	Harvest date, drying surface, threshing method and moisture.	Lot eligible for grading.
Grade gate	Moisture, foreign matter, damaged and shrivelled grain.	Grade A/B/C certificate.
Dispatch gate	Bag ID, warehouse, buyer and transport record.	QR verification active for buyer.

Each gate should be time-stamped. If a lot fails a gate, it can be re-cleaned, downgraded, diverted to feed/industrial use or held for further testing instead of entering premium food-grade supply.

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17. BAJARI VALUE ADDITION & MARKET PRODUCTS

17.1 Traditional Food Products

- Bajari bhakri, rotla, roti and village flour.
- Pearl millet porridge, khichdi, rabadi, lapsi and malted drinks.
- Roasted grain mixes and seasonal rural snacks.

17.2 Modern Health & Value-Added Products

Product	Processing Requirement	Market Position
Bajari flour	Cleaning, dehulling optional, milling, freshness control.	Daily-use gluten-free flour.
Multigrain atta blend	Controlled blending with wheat/jowar/pulses.	Urban health and family nutrition market.
Ready mix bhakri/roti flour	Fine milling, anti-rancidity packaging, recipe standardisation.	Convenience product.
Extruded snacks	Flour blending, extrusion and seasoning.	Better-for-you snack segment.
Malted Bajari drink	Germination, drying, roasting and milling.	Children and sports nutrition formulations.

17.3 Industrial Uses

- Feed ingredient for poultry, cattle and small ruminants when food-grade premium is not achieved.
- Starch and fermentation uses in local processing industries.
- Composite flour in bakery, noodles, breakfast mixes and institutional nutrition products.

17.2 Modern Health & Value-Added Products — Processing Controls

Product Line	Critical Control	Reason
Fresh Bajari flour	Mill in small batches and pack quickly.	Pearl millet fat content can reduce shelf life if flour is stored poorly.
Ready mixes	Standardise particle size and recipe ratio.	Improves cooking performance and consumer repeat purchase.
Malted products	Control germination time and drying temperature.	Prevents off-odour and microbial risk.
Extruded snacks	Use consistent moisture and flour blend.	Maintains expansion, texture and flavour.
Institutional foods	Retain assay certificate and lot traceability.	Supports procurement audit and nutrition claims.

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18. GOVERNMENT SCHEMES & POLICY SUPPORT

18.1 Key National Schemes for Millet Farmers

Scheme / Institution	Support Relevance
National Food Security Mission / Nutri Cereals	Promotes improved seed, demonstrations and extension for millets.
PMFME Scheme	Supports micro food-processing enterprises including Bajari flour and value-added products.
FSSAI Millet Standards	Quality parameters for safe millet marketing and processing.
AGMARK / DMI Grading Rules	Grade designation and quality standards for pearl millet trade.
e-NAM	Assaying and tradable quality parameters for transparent market transactions.

18.2 Maharashtra State Support

- State agriculture department advisories on district-specific seed, sowing and fertilizer management.
- Krishi Vigyan Kendra demonstrations for improved Bajari varieties, intercropping and dryland practices.
- FPO aggregation support for cleaning, grading, storage and market linkage.
- Millet promotion activities through nutrition campaigns, food festivals and local procurement opportunities.

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19. QUICK REFERENCE SUMMARY TABLE

Parameter	Bajari Recommendation
Crop family	Poaceae; pearl millet / Bajari
Botanical name	Cenchrus americanus (syn. Pennisetum glaucum)
Main Maharashtra districts	Dhule, Nandurbar, Nashik, Akola, Beed, Ahilyanagar, Chhatrapati Sambhajinagar
Season	Kharif; summer only under irrigation and local advisories
Sowing window	Mid-June to mid-July after effective monsoon rainfall
Seed rate	3–4 kg/ha line sowing; 4–5 kg/ha in risk-prone fields
Spacing	45 × 10–15 cm in normal rainfall zones
Soil	Well-drained sandy loam to medium black soil
Rainfall	350–600 mm; drought-tolerant but yield responds to moisture
Critical irrigation	Booting/flowering and grain filling
NPK	40–80 kg N/ha with P and K based on soil test
Major diseases	Downy mildew, blast, ergot and smut
Harvest	80–95 DAS, after earheads dry and grains harden
Grade A moisture	≤ 12% with clean, bold, uniform grain
Traceability must capture	Seed lot, plot, sowing, inputs, harvest, moisture, grade and bag IDs

— END OF REPORT —